CASE STUDY
AI in Network Security Project

Project Sponsor
Dr. Alice Ren (Chief Information Officer)
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Project Manager
Lead Systems Analyst (AI Research Division)

Project Customer
CyberGuard Solutions – Security Operations Centre (SOC) and Enterprise Clients

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Executive Summary
CyberGuard Solutions is a scheme the Australian cyber security firm is working on, in order to safeguard computer networks, to look at dangers and to locate flaws in security within systems. Because cyber-attacks are getting more sophisticated – notably attacks such as Advanced Persistent Threats and zero-day exploits a great number of organisations are employing Artificial Intelligence to strengthen their security and to recognise threats more easily (Stallings, 2018).

The intention of this project is to build an AI-Driven Network Anomaly Discovery System (NADS). The system will observe network movement, in real time, and discover odd behaviours using machine learning, and it will automatically hand over warnings to security personnel if a possible danger is spotted. Machine learning is often put to use in up-to-date intrusion detection systems since it is able to examine big quantities of network data and to spot trends which might be pointing to new or unknown cyber-attacks (Sommer & Paxson, 2010).

The project is to last for five months with an anticipated costing of $450,000. The progression will adhere to an Agile-Scrum approach; letting the team to work in short stages of development, to obtain constant comments from those with a stake in the project, and to little by little refine the system all through the length of the project (Schwaber & Sutherland, 2020).

Project Description
This project is about creating and putting in place an AI-based Network Anomaly Detection System or NADS which employs machine learning methods to look at what's happening in network traffic and find unusual activity (Sommer & Paxson, 2010).
The system is going to watch network traffic as it happens and give danger ratings to things that seem suspicious. When it finds something not normal, it will automatically make warnings and send them to the Security Operations Centre, the SOC, so they can deal with any security events.
The people working on the project are going to use an Agile Scrum approach; this allows for development in steps, with sprints, and also regular meetings to get views from people who have an interest in the project (Schwaber & Sutherland, 2020).

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Initial Risks

Risk ID	Risk Statement	Probability	Impact	Score	Strategy	Response
1	Resistance from employees in traditional security divisions	Medium	High	High	Mitigation	Increase communication and training
2	AI model accuracy may be insufficient during early stages	Medium	High	High	Mitigation	Use expert validation and model testing
3	Project delays due to complex machine learning development	Medium	Medium	Medium	Mitigation	Agile sprint planning and monitoring
4	Budget overruns due to external consultants	Low	Medium	Medium	Control	Careful cost monitoring
5	Integration challenges with existing network systems	Medium	Medium	Medium	Mitigation	Early infrastructure testing

Initial Stakeholders

Name	Interest	Impact	Strategies
Dr. Alice Ren (CIO)	Strategic success of AI security platform	High	Regular executive updates
Michael Tan (Chief Security Architect)	Technical performance and architecture	High	Weekly technical reviews
Jordan Blake (IT Infrastructure Lead)	System integration with network infrastructure	Medium	Collaboration with the engineering team
Dr. Sara Khalid (Head of Security Operations)	Operational effectiveness of threat detection	High	Continuous SOC feedback
Scrum Development Team	System design and implementation	High	Agile sprint meetings
Security Operations Centre (SOC)	End users of the system	High	User demonstrations and feedback

Output Of Project

- First release of an AI-based anomaly detection tool (MVP) for networks
- Threat detection engine utilizes machine learning technology.
- Constantly observing traffic on a live network.
- Automatic generation of alerts/triage of incidents.
- Outcome of validation of AI model.
- Outcomes of the security test and penetration test.

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Summary Milestones

Milestone	Due Date
Project Initiation and Planning	Month 1
System Architecture and Data Preparation	Month 2
AI Model Development	Month 3
Alpha Version Release	Month 4
MVP System Completion	Month 5

Summary Budget and Feasibility
The estimated project budget is **$450,000**, which includes personnel costs, external consultants, and infrastructure resources:

Project Manager: $150/hour × 2 days/week × 5 months
Technical Team: 6 team members × $95/hour

Outsourced specialists
• Consultant who validates AI models
• Pro External Penetration Test consultant. CyberGuard is growing strongly and there is a growing market demand for AI-based cybersecurity products and services so the project will be financially feasible.

Project Objectives and Success Criteria
Scope
Create and launch a minimal viable product of the AI-Powered Network Anomaly Detection System with features for real-time monitoring, detection, and automatic alerts.

Time
Finish the project in five months.

Cost
Completed the project within budget ($450,000)

Quality
Design a system that can identify anomalous behavior of network traffic with low false positives and is acceptable as per the enterprise security standards.

Other
Strengthen Internal Buy-In For AI Technologies. Support CyberGuard's plan of action to more towards AI system.

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Acceptance Criteria
The project will be accepted
• MVP system was delivered on time and within budget.
• Anomaly detection in real-time works as intended.
• Automated notifications produced and forwarded to SOC.
• AI models work well for detection.
• Security and penetration test.

Project Approval Requirements
The SSC must successfully undergo system testing, validation and acceptance by the Security Operations Centre for project approval.

Project Manager
Lead Systems Analyst – AI Research Division
CyberGuard Solutions

Approvals

Project Manager Signature	Sponsor or Originator Signature
Project Manager Name	Sponsor or Originator Name
Date	Date

References

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